

Sequential Decision-Making under Uncertainty

SCIE1200 Guest Lecture
25 Aug 2026

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an academic...



teaching



proving



hacking



supervising some very intelligent people

cartoons generated by AI

big picture #1 (my work)



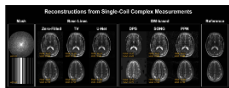
autonomous driving



routing behavior analysis



fishery stock assessment



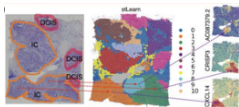
MRI reconstruction

theoretically grounded practical
algorithms
for
learning & decision-making

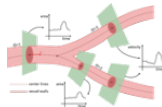


AAGI
Analytics for
the Australian
Grains Industry

agriculture analytics

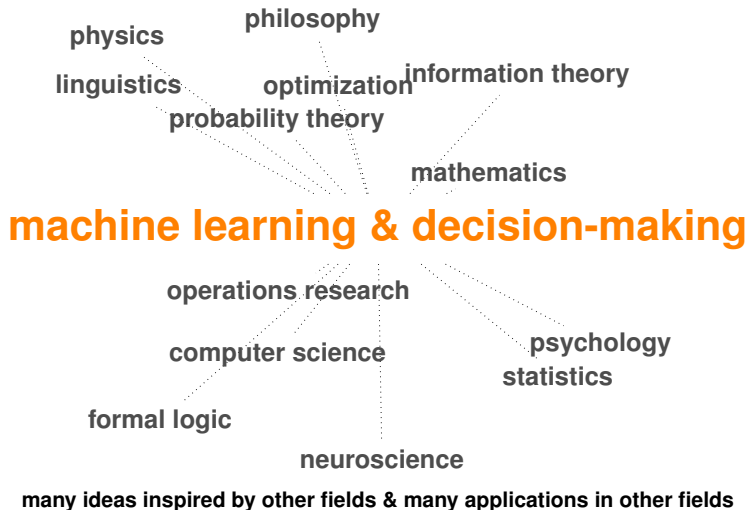


spatial transcriptomics



haemodynamics modeling

big picture #2 (synergy)





“Mama always said that sequential decision making under uncertainty life was like a box of chocolates.
You never know what you’re gonna get.”

Forrest Gump



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message#1. optimal decisions are often elusive (analytically and computationally)

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message#1. optimal decisions are often elusive (analytically and computationally)
message#2. good decisions are often computable (via iterative methods)

Teaser #1. Coin Flipping



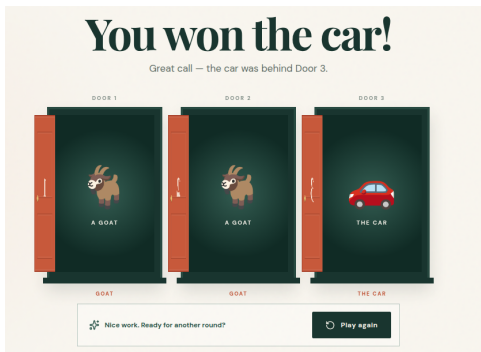
I have a biased coin with the probability of getting heads being 0.7.

I will toss the coin, and you will guess whether it lands heads or tails.

You get \$2 if you are right, otherwise you will pay me \$2.

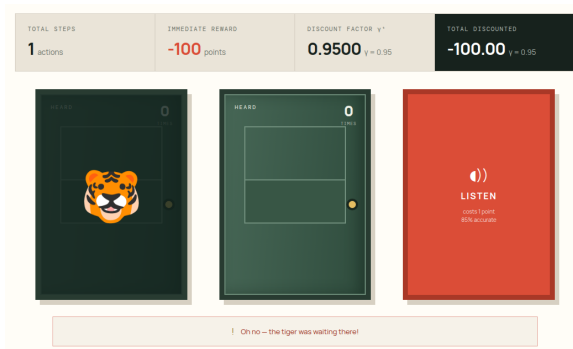
What will you guess?

Teaser #2. Monty Hall Problem



<https://huggingface.co/spaces/nanye/montyhall>

Teaser #3. Tiger

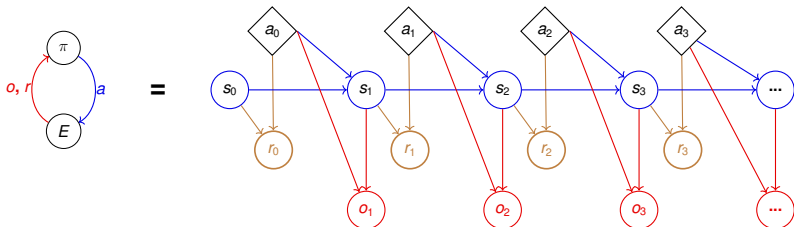


<https://huggingface.co/spaces/nanye/tiger>

summing up: decision-making can be hard due to

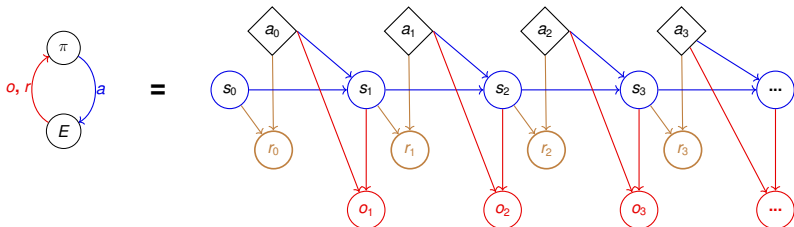
- the sequential nature
- uncertainty
- partial observability

General framework



an agent π interacting with an environment E

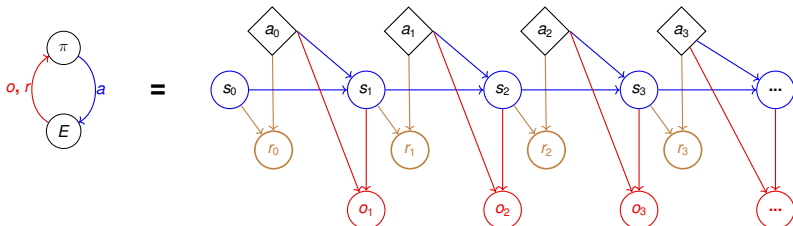
General framework



an agent π interacting with an environment E

at each time step, agent executes an action a , receives an observation o and a reward r , and environment state evolves

General framework

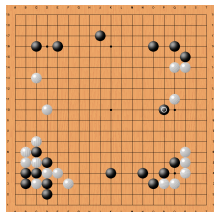


an agent π interacting with an environment E

at each time step, agent executes an action a , receives an observation o and a reward r , and environment state evolves

Example. Playing Go against an opponent

- state s_t : board position
- action a_t : a valid move
- observation o_t : s_t
- reward r_t : 0, except +1 for win and -1 for loss
- transition from s_t to s_{t+1} : determined by the opponent



Iterative Refinement

Exact analytical solutions are rarely available.

Many solution methods are computational and they iteratively refine a solution.

An iterative method can be written in the form $x_{t+1} = H_t(x_t)$

- x is the solution
 - this may be a policy or its value or both
 - common representations: tables, parametric functions (e.g., neural networks), trees
- refinement operator H_t uses model/interactions
 - this may be deterministic or stochastic
 - this may be defined explicitly or implicitly

idea #1. optimism in the face of uncertainty

Multi-armed Bandits (MABs)

What's the best sequence of pulls for K bandits (slot machines)?



Source: [Wikipedia](#)

Assumption: each pull gives a independent random reward $r_t \in [0, 1]$ sampled from a fixed distribution associated with the arm.

Objective: We would like to play to minimize the expected regret

$$R_T = T\mu_* - \mathbb{E}\left[\sum_{t=1}^T r_t\right],$$

where T is the number of pulls, $\mu_* = \max_i \mu_i$, and r_t is the reward at time step t .

Lower bound: R_T is at least of the order $O(\ln T)$ (Lai and Robbins, 1985).

UCB (Upper Confidence Interval): for each time step t , play machine j with maximum

$$\bar{x}_j + \sqrt{\frac{2 \ln t}{n_j}},$$

where $\bar{x}_j$ =average reward for machine j , and n_j = number of plays for machine j .

Theoretical guarantee: UCB achieves an optimal regret of $O(\ln T)$.

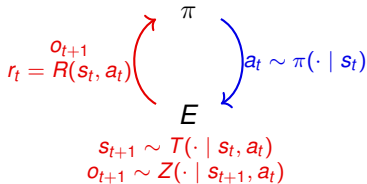
idea #2. sequential decision-making as tree search

POMDPs

Partially Observable Markov Decision Processes

POMDP $\mathcal{M} = (b_0, S, A, O, T, Z, R, \gamma)$

- initial state distribution b_0
- state/action/observation space $S/A/O$
- transition model $T(s' | s, a)$
- observation function $Z(o | s', a)$
- reward function $R(s, a)$
- discount factor $\gamma \in [0, 1]$



Planning: given $\mathcal{M}$, find an optimal policy $\pi : \mathcal{B} \rightarrow A$ such that

$$V_\pi(b) \geq V_{\pi'}(b) \text{ for any belief } b \text{ and any policy } \pi' : \mathcal{B} \rightarrow A$$

where $V_\pi(b) = \mathbb{E}(\sum_{t=0}^{\infty} \gamma^t r_t | s_0 \sim b, \pi)$ is the value of π starting from initial belief b , and $\mathcal{B}$ is the set of beliefs (probability distributions on S).

Algorithms are scaling up

- 1990's: tens of states is challenging.
- 2000's: offline point-based methods
PBVI (thousands of states), SARSOP (hundreds of thousands of states)
- after 2010: online belief tree search
POMCP, GPS-ABT, DESPOT, POMCPOW, ADVT, LCEOPT, ...

Offline algorithms plan for **all** beliefs

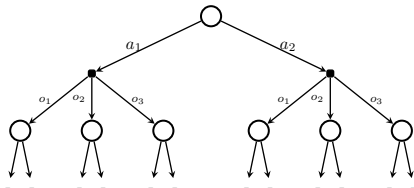
- Plan first, execute later.
- Planning is often slow, but execution is often fast.
- Generally do not exploit the local structure in the search space.

Online algorithms plan for **current** belief

- Interleave planning and execution.
- Possibility of benefiting from local structure in the search space.

Belief-tree Search

- A belief tree encapsulates all possible strategies

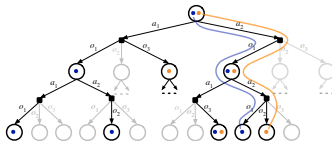


- root node is the initial belief
- following an action a from a belief node b and observing o leads to a belief node representing the updated belief
- Naive belief tree search
 - truncate the tree to a finite depth D (#beliefs = $O(|A|^D |O|^D)$)
 - estimate values of leaf beliefs
 - compute action values in a bottom-up manner
- Many online algorithms construct an iteratively refined approximation to the belief tree

Improved Tree-based Methods

DESPOT

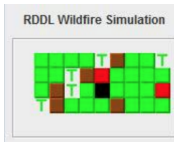
- An anytime online Monte Carlo algorithm for POMDPs with discrete actions and observations.
- It searches a sparse approximation to the belief tree.



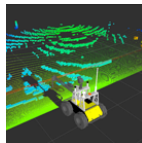
- Provably near-optimal.
- Good empirical performance



self-driving



IPPC'14 winner



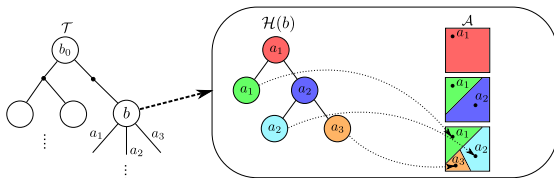
HRATC'15 winner

Adaptively discretized trees for continuous actions

- Robotic tasks often involve continuous actions



- ADVT



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